

Expanded Form, Comparing, and Rounding

Writing a number in expanded form, saying what each digit is worth, comparing and ordering multi-digit numbers, and rounding to a named place

Directions.

- A digit's value depends on where it sits. The same digit is worth ten times as much in the place to its left.
- To compare, start at the *largest* place, not the largest digit. Move right only when the digits match.
- To round, look at the digit *one place to the right* of the place you are rounding to. Five or more rounds up; four or less stays.
- Say the place out loud before you answer — ones, tens, hundreds, thousands, ten thousands. Naming the place is half of every problem here.

1. Write 6382 in expanded form.

Answer: _____

2. In the number 47308, what is the value of the digit 7?

Answer: _____

3. Round 6382 to the nearest hundred. Circle the digit you looked at to decide.

Answer: _____

4. Write $<$ or $>$ between 47308 and 47380. Name the place that decided it.

Answer: _____

5. A number is written in expanded form as

$$40000 + 3000 + 60 + 5.$$

Write the number in standard form. Be careful: one place has nothing in the expanded form, so it needs a placeholder.

Answer: _____

- 6.** Round 47308 to the nearest thousand. Name the digit you looked at.

Answer: _____

- 7.** Kai wrote $8090 > 8900$. He said, “the first one has a 9 in it and so does the second, but the first one also has an 8 at the front and a 9, so it must be bigger.”

Write the correct comparison, and say in one sentence which place Kai should have compared first.

Answer: _____

- 8.** Look at the number 55505.

The 5 in the ten-thousands place is worth 50000. The 5 in the hundreds place is worth 500. How many times greater is the first 5 than the second 5?

Answer: _____

- 9.** Round 25649 to the nearest thousand.

Answer: _____

- 10.** Put these three numbers in order from least to greatest:

72405 72045 70425

All three use the same five digits, so compare place by place from the left.

Answer: _____

11. Round 8950 to the nearest hundred.

Something unusual happens here. After you round, explain in one sentence why the thousands digit changed too.

Answer: _____

12. Challenge. A stadium counted 38472 people on Friday and 38247 people on Saturday.

(a) Write Friday's attendance in expanded form.

Answer: _____

(b) Which day had more people? Write the comparison using $<$ or $>$.

Answer: _____

(c) Round each attendance to the nearest thousand.

Answer: _____

(d) The two rounded numbers came out the same. Write one or two sentences explaining why the rounded numbers cannot settle which day had more people, and say which form you would use instead.